

SISEN: Teaching and Integration Guide

Security-Informed Safety

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Purpose of this Guide

This guide explains how SISEN can be used and integrated into teaching. It is intended for lecturers, module leaders, laboratory tutors, and curriculum designers who are deciding how the lecture materials, practical scenarios, activity guidance, and hazard-analysis resources should be used together.

SISEN introduces a security-informed safety perspective for wireless, IoT, and cyber-physical systems. Students observe normal operation, examine communication and telemetry, perform a bounded attack or disruption within the SISEN environment, collect evidence, and connect the resulting technical effect to a possible safety hazard and mitigation strategy.

1 What the SISEN Resource Pack Contains

The resource pack contains:

- **eight lecture sets**, covering security and safety fundamentals, security-informed safety, hazard analysis and assurance, IoT and cyber-physical systems, wireless threats, safety-critical IoT, resilience and mitigation, and applied case studies;
- **three practical scenario guides**, covering Medical IoT, Smart Building, and 6LoWPAN;
- the **Attack, Observation, and Disruption Guide**, which provides the bounded technical activities, capture procedures, observable effects, and safety relevance;
- the **SISEN Student Guide**, which provides common installation, setup, and working-practice guidance;
- the **Student Activity, Evidence and Hazard Analysis Sheet**, which supports evidence collection and structured security-to-safety reasoning; and
- the **Instructor Guide**, which provides detailed delivery, preparation, troubleshooting, and assessment guidance.

The materials are available through the SISEN repository and the SISEN project site.

2 CyBOK Alignment

SISEN supports teaching across several CyBOK knowledge areas, including introductory concepts, risk management and governance, adversarial behaviours, authentication and authorisation, network security, cyber-physical systems security, and physical-layer and telecommunications security.

The detailed lecture-level and section-level mapping is provided separately within the SISEN teaching materials. Instructors may therefore use the complete sequence or select only the material relevant to a particular module or learning outcome.

3 Ways to Integrate SISEN

SISEN can be used in several formats.

- **Single practical session:** Use one scenario to introduce the relationship between a security event, an operational effect, a safety hazard, and a mitigation.
- **Short teaching block:** Combine selected lectures with one or more practical scenarios over two or three sessions.
- **Module topic:** Use the lectures and labs as a focused block within cyber security, networking, IoT, cyber-physical systems, wireless security, or safety and risk modules.
- **Assessment activity:** Ask students to submit selected evidence, a security-to-safety chain, a completed hazard analysis, and justified mitigation recommendations.
- **Advanced extension:** Ask students to compare scenarios, inspect packet captures, evaluate trust boundaries, or propose resilience improvements.

For introductory use, instructors may provide the exact activity commands and focus student effort on observation and explanation. For advanced use, students may be expected to interpret captures, distinguish failures across communication layers, and justify controls using both security and safety principles.

4 Selecting a Practical Scenario

- **Medical IoT** is suitable for teaching trustworthy and timely patient telemetry, gateway trust, message authenticity, freshness, and the consequences of false or missing clinical data.
- **Smart Building** is suitable for teaching wireless infrastructure, MQTT telemetry, environmental and occupancy monitoring, availability, and the effect of shared infrastructure failures.
- **6LoWPAN** is suitable for teaching constrained wireless networking, layered trust, border-router behaviour, IPv6 adaptation, UDP telemetry, and end-to-end security reasoning.

Each scenario can be delivered independently. The selected scenario should match the module topic, available time, technical level of the students, and the evidence or analysis expected from the activity.

5 Recommended Teaching Sequence

A typical SISEN activity follows this sequence:

1. introduce the relevant security and safety concepts;
2. ask students to start the selected scenario and establish normal behaviour;
3. capture or observe the relevant traffic, telemetry, dashboard state, or logs;
4. perform one bounded observation, attack, or disruption activity;
5. compare the system behaviour before, during, and after the activity;
6. identify the initiating security event, affected component, operational effect, possible hazard, and potential harm;
7. propose preventive, detective, fail-safe, resilience, and recovery controls; and
8. record residual risk and limitations.

6 Getting Started

Before delivery, the instructor should:

1. review the relevant lecture and practical materials;
2. select the scenario and bounded activity;
3. decide whether the activity is formative, summative, or demonstration-based;
4. test the selected scenario and activity using the Instructor Guide;
5. provide students with the Student Guide, relevant scenario guide, Attack, Observation, and Disruption Guide, and activity worksheet; and
6. provide any local assessment instructions, submission requirements, or marking criteria.

Detailed setup, delivery, troubleshooting, expected observations, and assessment guidance are provided in the SISEN Instructor Guide.